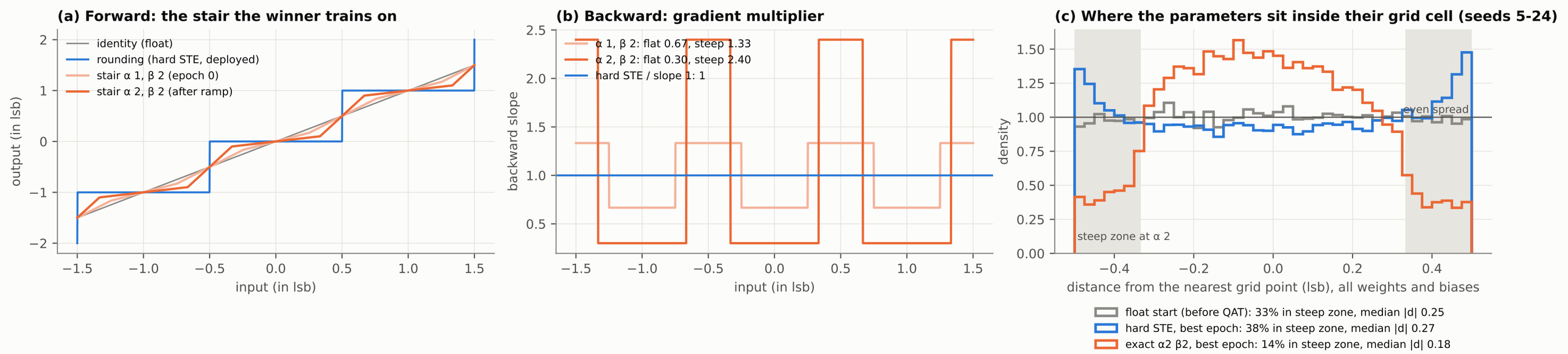
(a) Forward: the stair the winner trains on
identity (float)
rounding (hard STE, deployed)
stair α 1, β 2 (epoch 0)
stair α 2, β 2 (after ramp)
output (in lsb)
input (in lsb)
(b) Backward: gradient multiplier
α 1, β 2: flat 0.67, steep 1.33
α 2, β 2: flat 0.30, steep 2.40
hard STE / slope 1: 1
backward slope
input (in lsb)
(c) Where the parameters sit inside their grid cell (seeds 5-24)
density
steep zone at α 2
even spread
distance from the nearest grid point (lsb), all weights and biases
float start (before QAT): 33% in steep zone, median |d| 0.25
hard STE, best epoch: 38% in steep zone, median |d| 0.27
exact α2 β2, best epoch: 14% in steep zone, median |d| 0.18